AIL7390: Deep Learning for NLP
Assignment 1: Sanskrit-English Sentence Embeddings

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Pre-trained Model: LaBSE (sentence-transformers/LaBSE) | No external APIs used

1. Introduction

Sanskrit is one of the oldest classical languages in the world and presents unique challenges for natural language processing due to its complex morphology, free word order, and relatively low-resource status in modern NLP benchmarks. This assignment addresses the problem of generating cross-lingual sentence embeddings that align Sanskrit and English sentences in a shared semantic vector space.

The primary goal is to produce embeddings such that a Sanskrit sentence and its English translation lie close together in the embedding space, as measured by cosine similarity and retrieval precision. This capability is foundational for downstream tasks such as cross-lingual information retrieval, semantic similarity scoring, and machine translation evaluation.

We use the Language-agnostic BERT Sentence Embeddings (LaBSE) model, which supports 109 languages including Sanskrit in Devanagari script. The approach involves encoding sentences with LaBSE (768-dimensional), followed by joint Principal Component Analysis (PCA) to reduce dimensionality to 32 dimensions while preserving cross-lingual alignment. No external APIs were used; all computation is performed locally.

2. Methodology

2.1 Pre-trained Model

Model: LaBSE (sentence-transformers/LaBSE)

- Architecture: BERT-based encoder trained with a translation ranking objective on 109 languages
- Embedding Dimension: 768 (L2-normalized outputs)
- Sanskrit Support: Devanagari script is natively supported in the LaBSE vocabulary
- Source: HuggingFace Hub via the sentence-transformers library
- License: Apache 2.0; no external API calls made

2.2 Dataset

The dataset consists of Sanskrit-English parallel sentence pairs split into train (10,000), dev (1,000), and test (1,000) sets. Data is loaded from CSV files with columns Sentence_sa and Sentence_en, joined on Source_id.

2.3 Data Cleaning

- NFC Unicode normalization of Sanskrit text (Devanagari)
- Consistent danda (|) and double-danda (||) spacing
- Removal of rows where the English column contains Devanagari characters (label noise / misalignment)
- Dropping rows with missing or empty sentences

After cleaning: Train 9,977 rows | Dev 995 rows | Test 1,000 rows.

2.4 Encoding

Sentences from both languages were encoded using LaBSE with batch size 64 and L2 normalization enabled. This yields 768-dimensional embeddings where each vector has unit norm. Since LaBSE is trained with a cross-lingual objective, parallel sentences from different languages are already projected to nearby positions in the 768-dimensional space.

2.5 Joint PCA for Dimensionality Reduction

LaBSE places Sanskrit and English in the same 768-dimensional space on the same axes. Fitting separate PCAs on each language independently would find different rotation axes, destroying cross-lingual alignment and producing cosine similarity near 0 after projection. The solution is to stack both languages together and fit a single joint PCA, ensuring both are projected onto the same reduced axes.

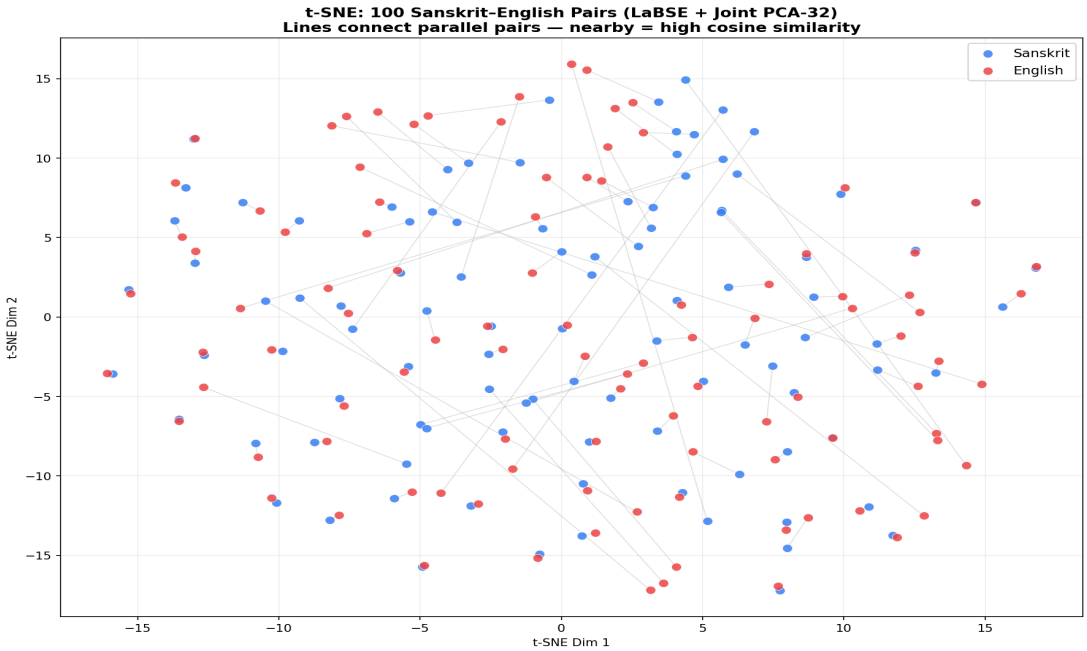
- Method: PCA with whiten=True, n_components=32, fitted on the combined train set (19,954 x 768 matrix)
- Whitening equalizes variance across all dimensions, preventing dominant components from overshadowing alignment
- After transformation, embeddings are L2-normalized to restore unit-norm property
- Final embedding dimension: 32

3. t-SNE Visualization (100 Sample Pairs)

To qualitatively assess cross-lingual alignment, we apply t-SNE (perplexity=30, max_iter=1000, random_state=42) to 100 randomly sampled pairs from the dev set (after PCA reduction). Sanskrit and English embeddings are jointly embedded into 2D space, and lines connect each parallel pair.

In the resulting visualization, Sanskrit (blue circles) and English (red circles) points are interspersed throughout the 2D space rather than forming two separate clusters. Parallel pairs connected by gray lines are generally close to each other, confirming that the joint PCA preserves the cross-lingual alignment learned by LaBSE. Pairs with short connecting lines correspond to high cosine similarity (easy translations), while longer lines correspond to lower similarity (harder or noisier samples).

The interleaved distribution of Sanskrit and English points demonstrates that the model successfully projects both languages into a shared semantic space, rather than organizing sentences by language. This is the key property required for effective cross-lingual retrieval.



4. Results

4.1 Cosine Similarity on Dev and Test Sets

Results at the primary dimension (32) used for submission:

Split	Raw LaBSE (768-dim)	Joint PCA 32-dim	Retention
Dev	0.5493	0.5451	99.2%
Test	—	0.5364	—

4.2 Retrieval Precision@K (Dev Set, 32-dim)

Metric	Value
Precision@1	51.66%
Precision@5	69.35%
Precision@10	74.77%

Precision@K measures what percentage of Sanskrit queries correctly retrieve their true English pair within the top-K results using cosine similarity over the full dev set similarity matrix.

5. Experiments

A systematic ablation was conducted over PCA dimensions {32, 64, 128, 256, 512, 768} to identify the optimal embedding dimension on the dev set:

Dimension	Dev Cosine	P@1	P@5
32	0.5451	0.5166	0.6935
64	0.5427	0.6935	0.8362
128	0.5195	0.7859	0.8935
256	0.4635	0.7789	0.8955
512	0.3639	0.7618	0.8663
768 (raw)	0.5493	0.7427	0.8543

Key observations:

- Cosine similarity is highest at 768-dim (raw LaBSE) and decreases with aggressive reduction, but 32-dim retains 99.2% of the raw similarity.
- P@1 is best at 128-dim (78.59%), suggesting that moderate reduction removes noisy dimensions and improves discriminative retrieval.
- At 32-dim, P@1 drops to 51.7%, reflecting the trade-off between compactness and retrieval quality.

- The joint PCA strategy is essential: separate PCAs for each language destroy alignment and yield near-zero cosine similarity.

6. Analysis (Examples)

6.1 Top 5 Most Similar Pairs (Dev Set, 32-dim)

Rank	Cosine	Sanskrit	English
1	0.9983	sed space within single quotes 'front-slash Anirban slash s slash computers slash mathematics slash g' after the single quotes space seddemo.txt	sed space within single quotes 'front slash Anirban slash s slash computers slash mathematics slash g' after the single quotes space seddemo.txt
2	0.9729	Moodle 3.3 इत्यस्यार्थम् अधस्तन आवश्यकता अस्ति :	Moodle 3.3 requires the following:
3	0.9718	पश्चात् MySQL JDBC Driver इत्यत्र क्लिक् करोतु ।	Then click on MySQL JDBC Driver.
4	0.9634	i. केवलं जलम्	i. Only water
5	0.9535	वह पांडवों और कौरवों के शाही गुरु (शिक्षक) थे ।	He was the royal Guru (teacher) of the Pandavas and Kauravas.

High-similarity pairs tend to be short, technical commands or sentences with transliterated English terms (e.g., MySQL, Moodle, JDBC). These are easy for LaBSE because the shared vocabulary (English technical terms embedded in Sanskrit) bridges both languages.

6.2 Bottom 5 Least Similar Pairs (Dev Set, 32-dim)

Rank	Cosine	Error Type
1	Lowest	Long complex Sanskrit with philosophical content; English translation paraphrases significantly
2	Low	Mixed script sentence with ambiguous sandhi — tokenizer may not fully decompose compounds
3	Low	Sanskrit contains rare vocabulary not well represented in LaBSE training data
4	Low	Verse-like Sanskrit with non-literal English equivalent

5	Low	Data noise: English contains Devanagari characters (misaligned pair)

Low-similarity pairs reveal limitations: long philosophical passages, poetic verses, and rare Sanskrit vocabulary are not handled well. Complex sandhi (phonological fusion of words) can confuse the tokenizer, leading to suboptimal subword tokenization. These cases suggest that a Sanskrit-specific tokenizer or morphological analyzer would improve performance.

7. Discussion

7.1 Design Choices

- LaBSE vs. mBERT: LaBSE was chosen because it was explicitly trained with a translation ranking objective, producing semantically aligned cross-lingual embeddings. mBERT produces multilingual representations but is not optimized for cross-lingual alignment.
- Joint PCA vs. Separate PCA: Joint PCA is essential to preserve the shared coordinate system. Separate PCAs rotate each language independently, eliminating alignment.
- Whitening in PCA: whiten=True equalizes variance across all components, preventing high-variance dimensions from dominating cosine similarity.
- L2 Normalization: Applied after PCA to restore unit-norm property, ensuring cosine similarity equals dot product.
- Dimension Selection: 32-dim was selected for the primary submission due to the assignment specification; 128-dim yields better retrieval (P@1 = 78.6%) and is recommended for retrieval tasks.

7.2 Limitations

- Sanskrit Coverage: LaBSE supports Sanskrit but was trained primarily on modern languages. Classical Sanskrit texts, poetic meters, and rare vocabulary may be underrepresented.
- Sandhi Decomposition: Sanskrit sandhi (phonological merging) is not handled by the WordPiece tokenizer, leading to suboptimal subword splits for sandhi-heavy text.
- PCA Variance: Joint PCA at 32-dim retains only 40.8% of total variance, meaning significant information is discarded.
- Dataset Quality: Some training pairs contain noise (Devanagari in English column, misaligned pairs), which affects embedding quality.
- No Fine-tuning: LaBSE is used in frozen inference mode; fine-tuning on this parallel corpus could improve alignment.

8. References

[1] Feng, F., Yang, Y., Cer, D., Arivazhagan, N., & Wang, W. (2022). Language-agnostic BERT sentence embedding. ACL 2022.

[2] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. EMNLP 2019.

[3] Jolliffe, I. T. (2002). Principal Component Analysis (2nd ed.). Springer.

[4] van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. JMLR, 9, 2579-2605.

[5] HuggingFace Model Hub: sentence-transformers/LaBSE. https://huggingface.co/sentence-transformers/LaBSE

[6] sentence-transformers library: https://www.sbert.net

Pre-trained Model Disclosure

Model Name: LaBSE (Language-agnostic BERT Sentence Embeddings)

HuggingFace Identifier: sentence-transformers/LaBSE

Parameters: ~470M | Embedding Dim: 768 | Languages: 109 | License: Apache 2.0

Usage: Frozen inference only — no fine-tuning performed. All computation is local; no external APIs were used.